

Advanced Functions & Equations



Function composition

- 1 Let $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. Find $f(g(2))$ and $g(f(2))$.
 - 2 Using the same f and g , find the formula for $(f \circ g)(x)$.
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Inverse functions

- 3 Find $f^{-1}(x)$ for $f(x) = 3x - 2$.
 - 4 Verify your answer above by computing $f(f^{-1}(5))$.
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Rational equations

- 5 Solve $\frac{2x - 1}{x + 3} = \frac{1}{2}$.
 - 6 Solve $\frac{3}{x - 1} + 2 = 5$.
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Factoring higher-degree polynomials

- 7 Given that $x = 1$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$, use the Factor Theorem to factor completely and find all roots.
 - 8 Verify that $x = -2$ is a root of $x^3 - 2x^2 - 5x + 6 = 0$ by direct substitution.
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Exponential and logarithmic equations

- 9 Solve $3^{x-1} = 81$.
 - 10 Solve $\log_3(x + 2) + \log_3(x - 2) = 2$, checking for extraneous roots.
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Multiple choice — easy

- 11 Let $f(x) = x - 9$ and $g(x) = -3x^2 + 7$. Find $f(g(0))$.
- a. 0
 - b. 7
 - c. -4
 - d. -2
- 12 Let $f(x) = -3x + 7$ and $g(x) = -2x^2 + 5$. Find $f(g(3))$.
- a. 46
 - b. 48
 - c. 44
 - d. -13
- 13 Let $f(x) = -2x - 6$ and $g(x) = -x^2 - 1$. Find $f(g(-4))$.
- a. 28
 - b. -17
 - c. 26
 - d. 30
- 14 Let $f(x) = -x + 9$ and $g(x) = 3x^2 + 1$. Find $f(g(-1))$.
- a. 3
 - b. 7
 - c. 4
 - d. 5

- 15 Let $f(x) = x + 7$ and $g(x) = x^2 - 8$. Find $f(g(-5))$.
- a. 26
 - b. 24
 - c. 17
 - d. 22
- 16 Let $f(x) = 4x - 8$ and $g(x) = 3x^2 + 6$. Find $f(g(6))$.
- a. 448
 - b. 450
 - c. 114
 - d. 446
- 17 Let $f(x) = 2x + 8$ and $g(x) = -3x^2 - 1$. Find $f(g(2))$.
- a. -20
 - b. -13
 - c. -18
 - d. -16
- 18 Let $f(x) = 4x + 4$ and $g(x) = -3x^2 - 7$. Find $f(g(-4))$.
- a. -218
 - b. -55
 - c. -214
 - d. -216

- 19 Let $f(x) = 3x - 1$ and $g(x) = 4x^2 + 1$. Find $f(g(-2))$.
- a. 52
 - b. 50
 - c. 17
 - d. 48
- 20 Let $f(x) = -3x + 3$ and $g(x) = -3x^2 + 3$. Find $f(g(1))$.
- a. 1
 - b. 0
 - c. 3
 - d. 5
- 21 Let $f(x) = 4x - 3$ and $g(x) = 2x^2 - 9$. Find $f(g(1))$.
- a. -7
 - b. -33
 - c. -29
 - d. -31
- 22 Let $f(x) = -4x - 9$ and $g(x) = -4x^2 - 3$. Find $f(g(-3))$.
- a. -39
 - b. 145
 - c. 147
 - d. 149

23 Let $f(x) = 2x - 1$ and $g(x) = 4x^2 + 2$. Find $f(g(1))$.

a. 13

b. 9

c. 11

d. 6

24 Let $f(x) = -4x - 5$ and $g(x) = 3x^2 - 3$. Find $f(g(0))$.

a. 7

b. -3

c. 9

d. 5

25 Let $f(x) = 2x + 3$ and $g(x) = 4x^2 - 8$. Find $f(g(5))$.

a. 187

b. 185

c. 189

d. 92

Multiple choice — medium

26 Solve $\frac{4x - 3}{x - 8} = 5$.

a. 38

b. 36

c. 37

d. -37

27 Solve $\frac{2x-4}{x+4} = -2$.

a. -1

b. 0

c. 1

d. -2

28 Solve $\frac{4x+5}{x-8} = -1$.

a. $-\frac{2}{5}$

b. $\frac{8}{5}$

c. $\frac{3}{5}$

d. $-\frac{3}{5}$

29 Solve $\frac{3x+4}{x-9} = -2$.

a. $-\frac{14}{5}$

b. $\frac{14}{5}$

c. $\frac{9}{5}$

d. $\frac{19}{5}$

30 Solve $\frac{3x-5}{x+6} = 1$.

a. $-\frac{11}{2}$

b. $\frac{13}{2}$

c. $\frac{11}{2}$

d. $\frac{9}{2}$

31 Solve $\frac{3x-6}{x+1} = -4$.

a. $\frac{9}{7}$

b. $\frac{2}{7}$

c. $-\frac{5}{7}$

d. $-\frac{2}{7}$

32 Solve $\frac{x+9}{x-2} = -4$.

a. $-\frac{1}{5}$

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. $-\frac{6}{5}$

33 Solve $\frac{-3x-7}{x+3} = 3$.

a. $\frac{8}{3}$

b. $-\frac{11}{3}$

c. $-\frac{5}{3}$

d. $-\frac{8}{3}$

34 Solve $\frac{-4x+5}{x-2} = -3$.

a. 0

b. 1

c. -2

d. -1

35 Solve $\frac{-2x+5}{x-9} = 1$.

a. $\frac{17}{3}$

b. $-\frac{14}{3}$

c. $\frac{11}{3}$

d. $\frac{14}{3}$

36 Solve $\frac{-x+2}{x-3} = 3$.

a. $\frac{7}{4}$

b. $\frac{11}{4}$

c. $-\frac{11}{4}$

d. $\frac{15}{4}$

37 Solve $\frac{3x+3}{x-7} = -2$.

a. $\frac{11}{5}$

b. $\frac{6}{5}$

c. $-\frac{11}{5}$

d. $\frac{16}{5}$

38 Solve $\frac{4x+6}{x-7} = -4$.

a. $\frac{11}{4}$

b. $-\frac{11}{4}$

c. $\frac{15}{4}$

d. $\frac{7}{4}$

39 Solve $\frac{-x+9}{x-1} = 2$.

a. $\frac{11}{3}$

b. $\frac{14}{3}$

c. $\frac{8}{3}$

d. $-\frac{11}{3}$

40 Solve $\frac{-3x+2}{x-6} = -2$.

a. -10

b. -9

c. 10

d. -11

Multiple choice — hard

41 Find $f^{-1}(x)$ for $f(x) = 2x - 5$.

a. $\frac{x+5}{2}$

b. $\frac{x+5}{-2}$

c. $2x - 5$

d. $\frac{x-5}{2}$

42 Find $f^{-1}(x)$ for $f(x) = 3x + 2$.

a. $3x + 2$

b. $\frac{x-2}{3}$

c. $\frac{x+2}{3}$

d. $\frac{x-2}{-3}$

43 Find $f^{-1}(x)$ for $f(x) = 4x + 7$.

a. $\frac{x-7}{-4}$

b. $4x + 7$

c. $\frac{x-7}{4}$

d. $\frac{x+7}{4}$

44 Find $f^{-1}(x)$ for $f(x) = -4x + 4$.

a. $-4x + 4$

b. $\frac{x+4}{-4}$

c. $\frac{x-4}{-4}$

d. $\frac{x-4}{4}$

45 Find $f^{-1}(x)$ for $f(x) = 2x - 8$.

a. $\frac{x+8}{2}$

b. $2x - 8$

c. $\frac{x+8}{-2}$

d. $\frac{x-8}{2}$

46 Given that $x = 4$ is a root of $x^3 - 13x - 12 = 0$, use the Factor Theorem to find the other two roots.

a. $(-3, -1)$

b. $(3, 1)$

c. $(-2, -1)$

d. $(-3, -2)$

- 47 Given that $x = -6$ is a root of $x^3 + 13x^2 + 54x + 72 = 0$, use the Factor Theorem to find the other two roots.
- a. $(-3, -3)$
 - b. $(-4, -4)$
 - c. $(4, 3)$
 - d. $(-4, -3)$
- 48 Given that $x = 3$ is a root of $x^3 - 10x^2 + 27x - 18 = 0$, use the Factor Theorem to find the other two roots.
- a. $(2, 6)$
 - b. $(1, 5)$
 - c. $(-1, -6)$
 - d. $(1, 6)$
- 49 Given that $x = -3$ is a root of $x^3 + 2x^2 - 33x - 90 = 0$, use the Factor Theorem to find the other two roots.
- a. $(-4, 6)$
 - b. $(5, -6)$
 - c. $(-5, 6)$
 - d. $(-5, 5)$
- 50 Given that $x = 2$ is a root of $x^3 + 6x^2 - 9x - 14 = 0$, use the Factor Theorem to find the other two roots.
- a. $(-6, -1)$
 - b. $(-7, -2)$
 - c. $(7, 1)$
 - d. $(-7, -1)$

51 Solve $2^{x+6} = 1$ for x .

a. -4

b. -7

c. -5

d. -6

52 Solve $2^{2x+1} = 2$ for x .

a. 1

b. 0

c. -1

d. 2

53 Solve $5^{2x+7} = 5$ for x .

a. -4

b. -2

c. -1

d. -3

54 Solve $3^{2x+6} = 1$ for x .

a. -1

b. -2

c. -4

d. -3

55 Solve $2^x = 8$ for x .

a. 5

b. 2

c. 3

d. 4